Placeholder

Anonymous ACL submission

Abstract

The emergence of Large Audio Language Models (LALMs) has broadened access to advanced and currently developing language technologies by supporting audio input alongside text. However, this convenience has primarily benefited speakers of high-resource languages, when this can be valuable to speakers of languages in low-resource settings, especially those where speech is the primary mode of communication. While recent research has focused on including low-resource languages by simply restricting training to a linear projector connecting a speech encoder and a large language model, as the decoder, the impact of the specialization of the large language model (LLM) used as the decoder remains unexplored in this setup. In this work, we investigate decoder specialization using a family of state-of-the-art (SOTA) small multilingual LLMs developed specifically to address the challenges of low-resource languages.

Extensive experiments across 12 languages, spanning a range of resource levels, including morphologically rich languages that remain challenging for automatic speech recognition (ASR) systems, reveal only minor differences in character error rate (CER) between the three regionally specialized decoders and their general-purpose counterpart. We release all checkpoints to support further investigation.

1 Introduction

Automatic speech recognition (ASR) models such as Whisper (Radford et al.) and Omnilingual ASR(team et al., 2025) have expanded multilingual speech recognition to a growing number of languages, including low-resource languages. Despite this progress, substantial work remains to address the breadth and diversity of low resource languages and dialects, which has motivated recent studies to investigate the failure modes of these systems in low-resource settings (Imam et al.; Kyslyi et al.; Kumar et al., 2026; Bharathi et al.; Azis et al., 2026). These limitations are important to address because language technologies today are increasingly multimodal, although text-based interfaces as the default in many settings and the speech modality is still developed primarily for high-resource languages. This is crucial given that, of the roughly 7,000 languages spoken worldwide, around 4,000 are estimated to have an established writing system, however, it does not correspond to widespread literacy or everyday use among speakers [1]. Considering this, Speech-first technologies may benefit speakers of languages in low resource settings by enabling them access to services that are better supported in high resource languages with current technologies. However developing end-to-end speech language systems remains computationally expensive and relies on abundant, high-quality annotated data.

One recent direction addressing these constraints is the SLAM-ASR architecture (Ma et al., 2024), which connects a pretrained speech encoder to a large language model (LLM) through a trainable linear projector, while keeping the pretrained models frozen. Restricting training to the projector provides a practical way to introduce a language in low-resource settings. Studies building on this work have investigated its robustness (Kumar et al., 2025), number of necessary audio hours (Fong et al., 2025), which encoder layers can be pruned while maintaining performance through fine-tuning of the LLM backbone (Kolluri et al., 2026), language-level and language-family-level training configurations (Zhang et al., 2026), and reducing prompt sensitivity with a learnable prompt projector (Burdisso et al., 2026). Although these prior studies have investigated several aspects of SLAM-ASR, the decoder selection has generally been guided by target-language support and state-

[1] Ethnologue, 28th edition https://www.ethnologue.com/

of-the-art (SOTA) multilingual coverage, [TODO: Add Examples of the LLMs they used]. To the best of our knowledge, however, the effect of decoder specialization within this architecture has not been systematically evaluated, leaving open whether specialized decoders provide an advantage over general-purpose models unexplored. More recently, the Tiny Aya family(Salamanca et al., 2026) introduced SOTA small multilingual LLMs that share a common multilingual foundation, with each regional model having undergone post-training to reinforce the linguistic clusters associated with its respective region.

Motivated by the growing study of failure modes in low-resource ASR and recent advances in small regionally specialized multilingual LLMs, we aim to contribute to the efforts to build linguistically informed language technologies. In this study we address the unexplored role of decoder specialization within the SLAM-ASR architecture by examining empirically comparing regional Tiny Aya decoders wth Tiny Aya Global across 10 languages , [TODO: Update]nine are from the three Tiny Aya regional clusters and one language unsupported by both and examining whether and to what extent specialization offers a strong starting point for low-resource languages when ASR data are scarce. [TODO: Introduce research questions(s) ?]

SLAM-ASR as a route into low-resource languages. The appeal of this architecture for low-resource settings is that almost nothing has to be trained. The encoder and the decoder are frozen and can be taken off the shelf, and the only optimized component is a projector of a few million parameters, so a language can be added with a few hours of transcribed audio and a single GPU rather than with a full ASR training pipeline. Fong et al. (2025) quantify where that route becomes competitive, finding that roughly 100–200 hours are needed before the framework matches a Whisper-only baseline, and that projector pretraining on a high-resource language substantially improves the low-resource end. Reviews of African ASR report the same shortage from the data side (Imam et al., 2025; Adjovi et al., 2026).

Most SLAM-ASR studies to date evaluate on a small set of well-resourced languages. Of the twelve languages we study, seven — French, Marathi, Urdu, Swahili, Hausa, Amharic and Kreol Seselwa — are not examined in the speech-LLM literature we survey, and Kreol Seselwa appears not to have been evaluated in this architecture at all. Reporting them together, under one recipe and one encoder, is part of what this study contributes.

2 Related Work

The projector-only recipe. Ma et al. (2024) showed that a single trainable linear projector between a frozen speech encoder and a frozen LLM reaches competitive English recognition, and subsequent work has probed the recipe from several directions: its robustness (Kumar et al., 2025), how much labelled audio it needs (Fong et al., 2025), how far the encoder can be pruned (Kolluri et al., 2026), how language-family structure affects it (Zhang et al., 2026), and its sensitivity to the training prompt (Burdisso et al., 2026). What these share is a fixed decoder: the LLM is chosen once, usually for its multilingual coverage, and then held constant. The decoder is treated as a component to be selected well, not as a variable to be studied.

Decoder choice where it has been varied. Fong et al. (2025) compare two decoders from different families, EuroLLM and Salamandra, and find a large gap at low resource — 14.0 against 33.6 WER at ten hours of Italian — which narrows as data grows. That establishes that the decoder can matter a great deal. It leaves open the question we ask here, because those decoders differ in scale, tokenizer and pretraining corpus at once. The Tiny Aya family isolates the variable: the variants share a base model and a tokenizer and differ only in the regional weighting of their post-training mix.

Low-resource speech recognition. Surveys of African ASR describe a persistent shortage of transcribed audio and of evaluation resources (Imam et al., 2025; Imam et al.), and recent work explores using ASR itself to bootstrap corpora for languages such as Fongbe and Hausa (Adjovi et al., 2026). Practitioner accounts for individual languages report the same constraint (Klejch et al., 2025). Against that background the projector-only recipe is attractive precisely because it asks for so little: no encoder training, no decoder training, and a component small enough to train on one GPU.

3 Methods

3.1 SLAM-ASR

[TODO: then remove specifics of encoder and decoder names and write them at the end.] Our

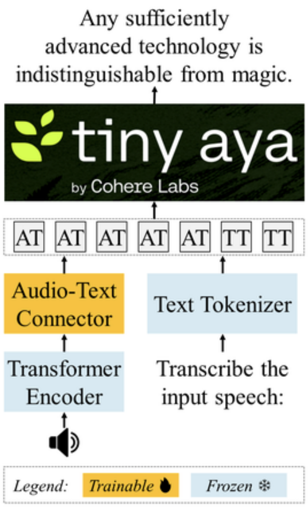

Figure 1: LisTAya

implementation follows the Encoder–Connector–Decoder architecture introduced in SLAM-ASR Ma et al. (2024) and is built upon the Qwen2-Audio codebase [2], as Illustrated in Figure 1

Following the original SLAM-ASR framework, only the Connector (projector) is optimized during training, while both the encoder and the decoder remain frozen. This design enables efficient adaptation by learning the alignment between speech representations and the language model embedding space while minimizing the number of trainable parameters.

3.2 Decoder Specialization

PH Note: Add details regarding the Tiny Aya Family, include a table.

To study the effect of decoder specialization on multilingual ASR, we employ models from the *TinyAya* family as the frozen decoder within the SLAM-ASR architecture.

TinyAya-Base is a pretrained 3.35B-parameter multilingual language model covering more than 70 languages. Built upon this foundation, *TinyAya-Global* provides balanced instruction-tuned performance across 67 languages, while three regionally specialized variants are available: *TinyAya-Earth* for languages from Africa and West Asia, *TinyAya-Fire* for South Asian languages, and *TinyAya-Water* for languages from the Asia-Pacific and Europe.

[TODO: Post training for each variant, tokenizer specialization]

[2] Qwen2-Audio codebase, `Qwen2-Audiocodebase`

All TinyAya variants share a common multilingual tokenizer, which is trained using a balanced weighting strategy that combines data-distribution and language-family information to achieve equitable representation of both high- and low-resource languages. Furthermore, TinyAya adopts a region-aware multilingual posttraining strategy in which high-quality instruction-tuning data is constructed through language-specific translation, prompt transformation, and Fusion-of-N teacher generation, enabling balanced multilingual instruction tuning across diverse language groups while improving coverage for low-resource languages.

4 Experimental Setup

4.1 Datasets

Our experiments use two multilingual speech corpora: World Speech (Asonitis et al., 2026) and FLEURS(Conneau et al., 2023). WorldSpeech primarily contains formal speech collected from diverse public sources such as parliamentary proceedings, international broadcasts, with smaller contributions from public-domain audiobooks. FLEURS contains read speech based on parallel sentences from the FLoRes-101 dataset, recorded were under relatively controlled conditions. We use subsets of train sets from WorldSpeech for training, excluding audio samples with durations of 30 seconds or longer. For evaluation, we use the validation sets from FLEURS when available and on the WorldSpeech test sets otherwise.

We select 12 languages for this study (Table 1). 11 are covered by the regional Tiny Aya models: Water (English, French, Spanish, and Indonesian), Fire (Hindi, Tamil, Marathi, and Urdu), and Earth (Swahili, Hausa, Amharic). Kreol Seselwa is an African language selected because it is unsupported by both backbone models. We include it to assess whether the trainable connector (projector) can effectively learn the alignment between speech representations and the language model, despite this lack of support.

4.2 Training Configuration

In our study, we use **Lis**tening **T**iny **A**ya (LisTAya; see Figure 1) to refer to models that use Whisper Medium as the fixed speech encoder, paired with different regional variants from the Tiny Aya family as decoders. Depending on the specialized regional decoder we refer to the resulting models as LisTAya-Water, LisTAya-Fire, LisTAya-Earth;

Language	WorldSpeech Train	FLEURS Test
English	en_us	en_us
Spanish	es_mx	es_419
French	fr_ca	fr_fr
Indonesian	in_in	id_id
Hindi	hi_in	hi_in
Marathi	mr_in	mr_in
Tamil	ta_in	ta_in
Urdu	ur_in + ur_pk	ur_pk
Swahili	sw_ke + sw_tz	sw_ke
Amharic	am_te	am_et
Hausa	ha_ng + ha_td	**ha_ng**
Kreol Seselwa	crs_sc	**crs_sc**

Table 1: Dataset overview. WorldSpeech training subsets are used to train the projector. Evaluation uses the FLEURS validation split; bold entries use the WorldSpeech test split. The + symbol denotes subsets pooled.

LisTAya-Global for the general-purpose variant.

Throughout all experiments, we train only the projector while keeping the encoder and decoder components frozen. We use the AdamW optimizer with a learning rate set at 1.5e-3, training steps 1, 000, batch size 8, gradient accumulation of 64 (an effective batch size of 512), and no weight decay. We retain the checkpoint after 1,000 steps, as well as an additional checkpoint achieving the lowest validation CER. All models were trained with one NVIDIA H200 GPU.

5 Results

5.1 Decoder specialization does not change what the model hears

Table 3 reports, for each language, the best CER of the region-matched decoder against the mean of the two mismatched regional decoders and against LisTAya-Global. Across the 11 languages covered by a Tiny Aya region, the matched decoder is **0.55 CER better** on average, with a 95% confidence interval of $[-3.89, 1.99]$ and a Wilcoxon signed-rank $p = 0.966$. Six of eleven languages favour the matched decoder and five do not.

This is a null, and its width is what makes it informative. The paired design's minimum detectable effect is **4.90 CER**, so effects smaller than that cannot be resolved here. We also measure the run-to-run floor directly rather than assuming it: over ten replicate pairs in which a cell was trained twice, the between-run standard deviation is **1.17 CER**, against a median within-run late-training standard deviation of 1.09. The observed effect is smaller than the noise between two runs of the same configuration.

5.2 The specialization is small in the data

A null invites the question of whether the manipulation was large enough to detect. We quantify it from the Tiny Aya report's own language-composition tables: for each language we take the share of the matched variant's post-training mix in that language, minus the mean share of the two mismatched variants. The median excess is **1.3 percentage points**.

Regional specialization in this model family is therefore a shift of about one percentage point of post-training data per language, not a change of training corpus. Read against that, the result is better stated as *specialization at this magnitude does not measurably change recognition accuracy*, which is a narrower and more useful claim than specialization being inert.

5.3 Training data volume predicts overfitting

While decoder choice does not predict accuracy, the amount of training audio predicts how a run behaves. Table 4 groups languages by resource tier and reports the rise in evaluation loss after its own minimum, and the fraction of the run elapsed at that minimum. Both are monotone across the four tiers: the lowest-resource cells reach their best evaluation loss about a quarter of the way through training and degrade over the remaining three quarters, while the highest-resource cells improve almost to the end.

5.4 Connector training versus off-the-shelf models

Table 5 compares each trained LisTAya checkpoint against the strongest off-the-shelf model we evaluate on the same test set, over all 43 cells. Training wins on 15 of them, and which 15 is the result: the benefit tracks how badly the off-the-shelf model was already doing (Spearman $\rho = -0.43$, $p = 0.004$). Splitting at the median baseline CER of 15.8, the projector wins 12 of the 22 cells at or above it and 3 of the 21 below, and the losses below are small (median +3.7 CER) while the wins above are not.

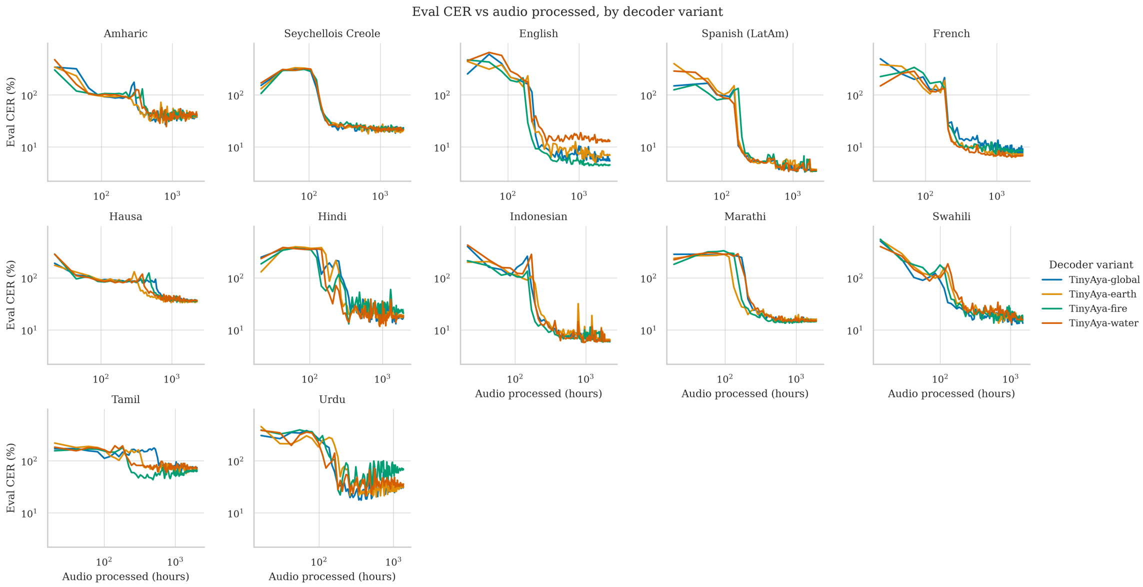

Figure 2: Character error rate against processed audio for every language and decoder variant. Each run is evaluated 101 times, so the curves show the whole trajectory rather than an endpoint. The four variants track one another closely within each language, while the languages themselves span two orders of magnitude of error.

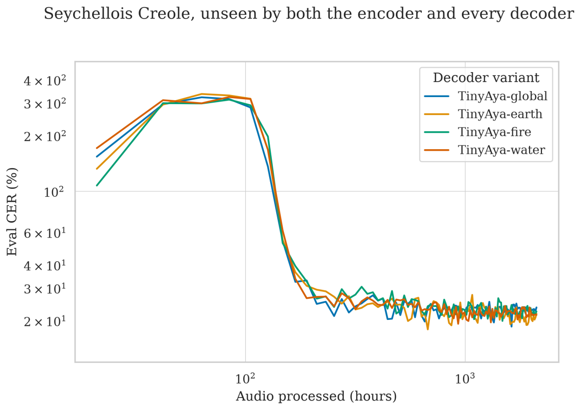

Figure 3: Kreol Seselwa, unsupported by both the encoder and every decoder. The four LisTAya variants converge to within 1.99 CER of one another.

Every large win is a language the frozen components barely cover: Amharic (-102.6 CER in-domain, -78.7 on FLEURS), Kreol Seselwa (-68.8), Hausa (-45.4) and Tanzanian Swahili (-32.0).

So training a projector does not make a system uniformly better. It converts unusable output into usable output where the encoder's coverage is thin, and gives up ground where a general-purpose model is already strong — which is the trade a practitioner adding a low-resource language is making.

5.5 A language neither component supports

Kreol Seselwa is supported by neither the Whisper encoder nor any Tiny Aya decoder, and it is the one cell where both frozen components operate outside their coverage. The four LisTAya variants span 17.92–19.91 CER on it, a range of 1.99 CER; adding the non-regional Tiny Aya base control widens that to 3.43. The projector therefore learns a usable alignment for a language absent from both pretraining corpora, and which regional variant it is paired with matters as little there as elsewhere.

5.6 Exposure explains the one language that disagrees

Table 6 gives the exposure figures per language. Ten of eleven behave as the region labels suggest, with the matched decoder having seen between 0.9 and 5.7 percentage points more of the target language than the mismatched ones.

English is the exception, and it is informative. English is a Water language, but the Fire mix contains 46.2% English against Water's 17.0%, so English is the one language whose *matched* decoder saw substantially *less* of it than the mismatched decoders did: an excess of -15.2 percentage points. English is also the single language where the matched decoder does worst, at $+7.39$ CER. Where the region label and the actual exposure disagree, the outcome follows the exposure

Language	**Corpus**	n	**Mean**	**SD**	**Min**	**Max**
Amharic	FLEURS	516	11.3	3.5	5.0	28.9
Amharic	WorldSpeech	468	14.7	8.8	3.0	30.0
Seychellois Creole	WorldSpeech	19162	15.0	2.8	10.0	20.0
English	FLEURS	647	9.9	3.6	3.6	29.3
English	WorldSpeech	35336	19.0	2.4	10.0	20.0
Spanish (LatAm)	FLEURS	908	12.3	3.7	4.9	30.3
Spanish (LatAm)	WorldSpeech	10839	15.0	2.8	2.1	20.0
French	FLEURS	676	10.4	3.7	3.5	33.5
French	WorldSpeech	10919	18.4	11.5	3.0	360.5
Hausa	FLEURS	621	19.4	8.7	6.1	78.9
Hausa	WorldSpeech	810	15.5	8.9	3.0	30.0
Hindi	FLEURS	418	11.6	4.4	3.8	31.4
Hindi	WorldSpeech	27474	13.7	9.2	1.0	30.0
Indonesian	FLEURS	687	12.4	4.5	4.4	41.3
Indonesian	WorldSpeech	5323	15.3	4.6	2.9	30.0
Marathi	FLEURS	1015	13.7	5.2	4.7	41.6
Marathi	WorldSpeech	3064	13.4	7.2	1.0	29.9
Swahili	FLEURS	487	14.2	4.4	4.6	29.2
Swahili	WorldSpeech	5352	8.6	6.2	1.2	30.0
Tamil	FLEURS	591	13.0	6.2	3.5	60.0
Tamil	WorldSpeech	466	13.9	7.2	2.0	29.9
Urdu	FLEURS	299	9.8	3.4	3.7	25.8
Urdu	WorldSpeech	1482	9.1	8.8	1.0	30.0

Table 2: Evaluation sets: number of utterances and audio duration in seconds, measured directly from the decoded audio at evaluation time.

rather than the label.

Across all eleven languages the correlation between excess exposure and the region-match effect is negative, as the account predicts, but not significant at this sample size (Pearson $r = -0.53$, $p = 0.09$; Spearman $\rho = -0.29$, $p = 0.38$).

5.7 Does the effect depend on how much data the connector had?

A natural hypothesis is that specialization should matter most where the projector has least to learn from. Figure 4 plots the region-match effect against training-stream size. An earlier, smaller version of this grid showed a strong monotone relationship; it did not survive the addition of languages. Over all eleven the rank correlation is $\rho = 0.48$ ($p = 0.13$), and dropping the extreme point leaves $\rho = 0.31$ ($p = 0.38$).

The confound is worth stating plainly rather than arguing away: training-stream size and baseline difficulty are strongly collinear here ($\rho = -0.82$, $p = 0.002$), because the languages with least audio are also the languages the encoder handles worst. A partial correlation controlling for baseline CER leaves $r = -0.13$ ($p = 0.72$) on eight residual degrees of freedom, which is inconclusive rather than null. Separating the two would need a within-language manipulation of training volume, which we do not have.

5.8 Learning speed separates the variants more than accuracy does

Endpoint accuracy is one number per run. Because every run is evaluated 101 times, we can also ask how quickly each decoder reaches a given error rate. Ranking the four variants by the processed audio needed to first come within $1.5\times$ of their own best CER, averaged over languages, gives LisTAya-Fire fastest (mean rank 2.00), then Water (2.08), Earth (2.75) and Global slowest (3.17).

That ordering is not the accuracy ordering, which runs Earth (2.17), Fire (2.50), Water and Global (2.67 each). How fast a projector converges and how well it ends up are behaving as separate axes. Both rankings place Global last, which is the only signal in this study pointing towards regional spe-

Language	Region	Match	Mism.	Global	Δ
Amharic	Earth	29.25	27.93	27.00	1.32
Tamil	Fire	43.63	58.33	61.21	-14.70
Urdu	Fire	22.24	20.81	17.64	1.44
Hausa	Earth	33.43	34.62	35.03	-1.19
Marathi	Fire	13.57	14.01	14.14	-0.44
English	Water	12.05	4.67	4.96	7.39
Spanish (LatAm)	Water	3.41	3.35	3.40	0.07
French	Water	6.36	6.79	6.61	-0.43
Hindi	Fire	13.11	12.05	13.18	1.06
Indonesian	Water	5.74	6.14	6.05	-0.40
Swahili	Earth	13.53	13.68	12.60	-0.15

Table 3: Best CER per language for the region-matched decoder, the mean of the two mismatched regional decoders, and LisTAya-Global. Δ is matched minus mismatched; negative favours the matched decoder. Ordered by resource tier.

Tier	Langs	Eval-loss rise	Frac. to best
very low	2	0.211	0.282
low	1	0.188	0.292
mid	2	0.041	0.627
high	7	0.003	0.816

Table 4: Overfitting by resource tier. Both columns are medians of per-language medians, so a tier with seven languages cannot outweigh one with two.

cialization mattering at all; with one seed per cell it should be read as suggestive rather than established.

5.9 Overfitting, not domain shift

Two quantities separate what a single error rate conflates. The generalisation gap, evaluation minus training loss, rises with both overfitting and distribution shift. The rise in evaluation loss after its own minimum rises only with overfitting, since it needs no comparison against the training distribution.

The tier result in Table 4 uses the second. The first does *not* distinguish our two evaluation domains: the generalisation gap is 0.184 for the ten cross-domain languages and 0.179 for the two in-domain ones. With two in-domain languages this comparison is underpowered in either direction, and we draw no conclusion from it. The evaluation-loss rise does differ, at 0.009 cross-domain against 0.001 in-domain, but both are close enough to zero that the tier contrast is the load-bearing result.

5.10 Off-the-shelf models on these languages

Table 7 reports the three off-the-shelf systems on every language. The spread is wide and tracks encoder coverage: Whisper-Medium's pretraining includes 438k hours of English and 17.8k of Spanish but only 289 hours of Marathi, 287 of Swahili, 32 of Amharic and 8 of Hausa, and the error rates follow. Several entries exceed 100 CER, which indicates degenerate output — repetition and hallucinated continuations — rather than ordinary substitution errors, and is the failure mode these models show on languages outside their coverage.

This is the context in which the projector result should be read. Training a projector on a few hours of in-language audio does not merely improve on these numbers, it changes the regime for languages where the off-the-shelf systems produce unusable output.

5.11 Accent and dialect transfer

WorldSpeech ships several country varieties per language, and the study trained on only some of them. Evaluating the rest costs inference alone and turns each in-domain number into a dialect-transfer axis. Twenty further varieties are evaluated by every model, of which 18 are genuinely held out — 8 Spanish, 7 English, 2 French and Sri Lankan Tamil — while Tanzanian Swahili and Indian Urdu were themselves part of their cells' training streams and are reported separately throughout.

The projector does not transfer across dialect. On its own training variety it beats the baseline on 9 of 12 cells, with a median gain of 3.0 CER. On genuinely held-out varieties of the same languages

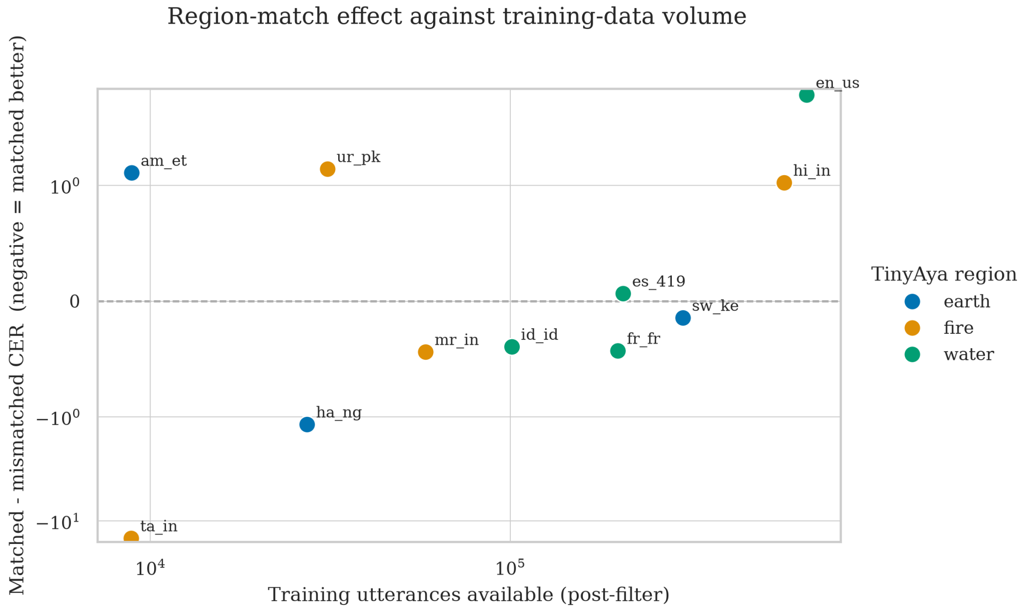

Figure 4: Region-match effect against training-stream size, one point per language. Negative favours the matched decoder. The vertical axis is symmetric-log so that Tamil, at -14.7 CER, does not compress the remaining ten languages into an unreadable band.

it wins *none* of the 18, at a median cost of 4.1 CER. The only two wins among the wider set are exactly the two varieties that were not held out, where it gains a median of 17.4 CER. A projector trained on one variety of a language is a system for that variety, and the apparent exceptions are seen data.

This also bears on whether a cheaper evaluation can stand in for a dialect sweep. Across the off-the-shelf models, a model's score on a language's in-domain test set predicts its score on held-out varieties of that language at $\rho = 0.42$ ($p = 0.001$, $n = 54$), and the association survives removing the language mean at $\rho = 0.30$ ($p = 0.026$). The FLEURS score predicts less well and does not survive it ($\rho = 0.20 \rightarrow 0.06$, $p = 0.66$). Most of any pooled correlation here is the language rather than the model: on the full variety set, including the two seen ones, in-domain appears to predict at $\rho = 0.53$ and falls to $\rho = 0.15$ once language is removed.

6 Discussion

What a practitioner should take from this. If the goal is to add a low-resource language to a speech interface, the decoder variant is not where the decision lies. Across eleven languages spanning four orders of magnitude of encoder pretraining, choosing the regionally matched Tiny Aya decoder over a mismatched one moves character error rate by less than the difference between two runs of the same configuration. What does move the result is whether a projector is trained at all, and how much in-language audio is available to train it.

Why the null is narrow, not empty. Two measurements bound it. The design resolves effects down to 4.90 CER, and the manipulation itself is about 1.3 percentage points of post-training data. A null over a treatment that small is not evidence that decoder specialization cannot matter; Fong et al. (2025) show that it can, between model families, by a wide margin. It is evidence about this family and this magnitude, which is the regime a practitioner choosing among released regional variants actually faces.

Where specialization would be expected to show. The exposure analysis suggests where to look next. The effect, insofar as one exists, tracks how much of the target language the decoder saw rather than which region it is labelled with. A decoder differing by tens of percentage points of in-language post-training data — as the regional variants do for English, by accident — is a much stronger manipulation than the regional labels provide, and English is the one cell where a large effect appears.

The frozen decoder is doing less work than its size suggests. Kreol Seselwa is supported by neither frozen component, and the projector still reaches error rates comparable to the low-resource end of the supported languages. Whatever the decoder contributes here, it is not primarily language-specific knowledge of the target language, which is consistent with the region labels making so little difference.

7 Conclusion

We asked whether the regional specialization of a frozen decoder changes what a SLAM-ASR model can hear. Across twelve languages and four Tiny Aya decoders under one recipe, it does not, at the magnitude these variants differ by: the region-matched decoder is 0.55 CER better on average with $p = 0.966$, against a design that resolves 4.90 CER and a between-run standard deviation of 1.17 CER measured from replicate pairs. The manipulation is about 1.3 percentage points of post-training data, and the one language whose exposure contradicts its region label behaves as exposure predicts.

Training the projector at all is what matters, though not uniformly. Across all 43 evaluated cells it beats the strongest off-the-shelf model on 15, and the benefit tracks the baseline's weakness ($\rho = -0.43$, $p = 0.004$): up to 102.6 CER where those models produce degenerate output, and a net cost where they are already strong. And how much in-language audio the projector sees predicts how the run behaves: overfitting falls monotonically across resource tiers, with the lowest-resource cells peaking a quarter of the way into training.

For practitioners adding a low-resource language, this is an encouraging result. The component that has to be chosen carefully is not the decoder variant; a general-purpose multilingual decoder suffices, and the effort is better spent on in-language audio. We release all checkpoints so that the comparison can be extended to other decoder families, where the differences are larger and, on the present evidence, more likely to matter.

Limitations

Most cells are trained once. Ten of the 52 runs have a replicate, which is what lets us state a between-run standard deviation of 1.17 CER, but the remaining cells carry no direct uncertainty estimate, and the design's minimum detectable effect of 4.90 CER is well above the effects that would be practically interesting. A negative result at this width bounds the effect rather than excluding it.

Ten of the twelve languages train on WorldSpeech and evaluate on FLEURS, so most reported numbers combine decoder choice with a domain shift from formal recorded speech to read speech. We report in-domain WorldSpeech test results alongside them where available, but the two are not separable in every cell.

The study measures transcription accuracy only. Whether decoder specialization affects downstream spoken-language understanding, where the decoder's own multilingual knowledge is more directly exercised, is not addressed here.

Finally, the comparison assumes the Tiny Aya variants differ only in post-training data. We take that from the model family's own documentation rather than verifying it independently.

Acknowledgments

PH

References

Mahounan Pericles Adjovi, Victor Olufemi, Roald Eiselen, and Prasenjit Mitra. 2026. From Speech to Text Corpora: Evaluating ASR-Based Data Acquisition for Low-Resource Fongbe and Hausa. arXiv:2606.22274.

Antonis Asonitis, Luca A. Lanzendörfer, Frédéric Berdoz, and Roger Wattenhofer. 2026. WorldSpeech: A Multilingual Speech Corpus from Around the World. *Preprint*, arXiv:2605.09167.

Huzain Azis, Muh Fatwah Fajriansyah M, Herdianti Darwis, Purnawansyah, Tasrif Hasanuddin, and Sugiarti. 2026. Development of a Low-Resource Automatic Speech Recognition System for the Makassar Dialect. In *2026 20th International Conference on Ubiquitous Information Management and Communication (IMCOM)*, pages 1–8.

B Bharathi, Bharathi Raja Chakravarthi, Shunmuga Priya Muthusamy Chinnan, S Saranya, and S Suhasini. Findings in Tamil Dialect Speech Recognition and Classification.

Sergio Burdisso, Esaú Villatoro-Tello, Shashi Kumar, Srikanth Madikeri, Andrés Carofilis, Pradeep Rangappa, Manjunath K E, Kadri Hacioglu, Petr Motlicek, and Andreas Stolcke. 2026. Reducing Prompt Sensitivity in LLM-Based Speech Recognition Through Learnable Projection. In *ICASSP 2026 - 2026 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, pages 16372–16376.

Alexis Conneau, Min Ma, Simran Khanuja, Yu Zhang, Vera Axelrod, Siddharth Dalmia, Jason Riesa, Clara

Rivera, and Ankur Bapna. 2023. FLEURS: FEW-Shot Learning Evaluation of Universal Representations of Speech. In *2022 IEEE Spoken Language Technology Workshop (SLT)*, pages 798–805.

Seraphina Fong, Marco Matassoni, and Alessio Brutti. 2025. Speech LLMs in Low-Resource Scenarios: Data Volume Requirements and the Impact of Pretraining on High-Resource Languages. In *Proc. Interspeech 2025*, pages 2003–2007.

Sukairaj Hafiz Imam, Tadesse Destaw Belay, Kedir Yassin Husse, Ibrahim Said Ahmad, Idris Abdulmumin, Hadiza Ali Umar, Muhammad Yahuza Bello, Joyce Nakatumba-Nabende, Seid Muhie Yimam, and Shamsuddeen Hassan Muhammad. 2025. Automatic Speech Recognition (ASR) for African Low-Resource Languages: A Systematic Literature Review. arXiv:2510.01145.

Sukairaj Hafiz Imam, Babangida Sani, Dawit Ketema Gete, Bedru Yimam Ahamed, Ibrahim Said Ahmad, Idris Abdulmumin, Seid Muhie Yimam, Muhammad Yahuza Bello, and Shamsuddeen Hassan Muhammad. Automatic Speech Recognition for African Low-Resource Languages: Challenges and Future Directions.

Ondřej Klejch, William Lamb, and Peter Bell. 2025. A Practitioner's Guide to Building ASR Models for Low-Resource Languages: A Case Study on Scottish Gaelic. In *Proc. Interspeech 2025*, pages 728–732.

Ganesh Pavan Kartikeya Bharadwaj Kolluri, Michael Kampouridis, and Ravi Shekhar. 2026. On the Role of Encoder Depth: Pruning Whisper and LoRA Fine-Tuning in SLAM-ASR. *Preprint*, arXiv:2603.27981.

Chowdam Venkata Kumar, Kumud Tripathi, and Pankaj Wasnik. 2026. Overcoming Decoder Inconsistencies in Whisper for Dravidian and Low-Resource Languages. *Preprint*, arXiv:2606.09535.

Shashi Kumar, Iuliia Thorbecke, Sergio Burdisso, Esaú Villatoro-Tello, Manjunath K. E, Kadri Hacioğlu, Pradeep Rangappa, Petr Motlicek, Aravind Ganapathiraju, and Andreas Stolcke. 2025. Performance evaluation of SLAM-ASR: The Good, the Bad, the Ugly, and the Way Forward. *Preprint*, arXiv:2411.03866.

Roman Kyslyi, Artem Orlovskyi, Pavlo Khomenko, Bohdan Onyshchenko, and Zakhar Guzii. Building ASR Resources for the Hutsul Dialect of Ukrainian.

Ziyang Ma, Guanrou Yang, Yifan Yang, Zhifu Gao, Jiaming Wang, Zhihao Du, Fan Yu, Qian Chen, Siqi Zheng, Shiliang Zhang, and Xie Chen. 2024. An Embarrassingly Simple Approach for LLM with Strong ASR Capacity. *Preprint*, arXiv:2402.08846.

Alec Radford, Jong Wook Kim, Tao Xu, Greg Brockman, Christine McLeavey, and Ilya Sutskever. Robust Speech Recognition via Large-Scale Weak Supervision.

Alejandro R. Salamanca, Diana Abagyan, Daniel D'souza, Ammar Khairi, David Mora, Saurabh Dash, Viraat Aryabumi, Sara Rajaee, Mehrnaz Mofakhami, Ananya Sahu, Thomas Euyang, Brittawnya Prince, Madeline Smith, Hangyu Lin, Acyr Locatelli, Sara Hooker, Tom Kocmi, Aidan Gomez, Ivan Zhang, and 7 others. 2026. Tiny Aya: Bridging Scale and Multilingual Depth. *Preprint*, arXiv:2603.11510.

Omnilingual ASR team, Gil Keren, Artyom Kozhevnikov, Yen Meng, Christophe Ropers, Matthew Setzler, Skyler Wang, Ife Adebara, Michael Auli, Can Balioglu, Kevin Chan, Chierh Cheng, Joe Chuang, Caley Droof, Mark Duppenthaler, Paul-Ambroise Duquenne, Alexander Erben, Cynthia Gao, Gabriel Mejia Gonzalez, and 14 others. 2025. Omnilingual ASR: Open-Source Multilingual Speech Recognition for 1600+ Languages. *Preprint*, arXiv:2511.09690.

Yuchen Zhang, Ravi Shekhar, and Haralambos Mouratidis. 2026. Language Family Matters: Evaluating LLM-Based ASR Across Linguistic Boundaries. *Preprint*, arXiv:2601.18899.

A Example Appendix

This is an appendix.

Language	Test set	Domain	Base.	LisTAya	Δ
Amharic	am_et	FLEURS	111.61	32.87	-78.74
Amharic	am_et	in-domain	133.07	30.45	-102.62
Seychellois Creole	crs_sc	in-domain	85.18	16.41	-68.77
English	en_us	FLEURS	2.11	5.54	3.43
English	en_au	in-domain	9.46	14.93	5.47
English	en_jm	in-domain	7.62	9.54	1.92
English	en_ke	in-domain	19.49	23.81	4.32
English	en_nz	in-domain	8.47	17.30	8.83
English	en_pk	in-domain	78.47	422.88	344.41
English	en_sl	in-domain	14.12	20.25	6.13
English	en_us	in-domain	9.95	9.44	-0.51
English	en_zm	in-domain	15.83	21.38	5.55
Spanish (LatAm)	es_419	FLEURS	1.34	3.14	1.80
Spanish (LatAm)	es_ar	in-domain	20.01	21.96	1.95
Spanish (LatAm)	es_cl	in-domain	19.59	23.07	3.48
Spanish (LatAm)	es_co	in-domain	11.79	15.16	3.37
Spanish (LatAm)	es_es	in-domain	7.43	13.01	5.58
Spanish (LatAm)	es_mx	in-domain	8.46	8.91	0.45
Spanish (LatAm)	es_pe	in-domain	6.55	10.39	3.84
Spanish (LatAm)	es_pr	in-domain	9.54	12.24	2.70
Spanish (LatAm)	es_py	in-domain	8.65	12.36	3.71
Spanish (LatAm)	es_uy	in-domain	20.69	23.48	2.79
French	fr_fr	FLEURS	2.10	7.47	5.37
French	fr_ca	in-domain	10.93	8.58	-2.35
French	fr_cd	in-domain	7.18	10.92	3.74
French	fr_ci	in-domain	7.77	13.21	5.44
Hausa	ha_ng	FLEURS	55.94	39.14	-16.80
Hausa	ha_td	in-domain	82.00	36.58	-45.42
Hindi	hi_in	FLEURS	5.74	13.73	7.99
Hindi	hi_in	in-domain	24.50	25.75	1.25
Indonesian	id_id	FLEURS	3.26	5.86	2.60
Indonesian	id_id	in-domain	13.89	10.85	-3.04
Marathi	mr_in	FLEURS	19.45	14.39	-5.06
Marathi	mr_in	in-domain	18.62	10.36	-8.26
Swahili	sw_ke	FLEURS	26.32	15.10	-11.22
Swahili	sw_ke	in-domain	18.63	14.52	-4.11
Swahili	sw_tz	in-domain	45.45	13.46	-31.99
Tamil	ta_in	FLEURS	16.22	48.29	32.07
Tamil	ta_in	in-domain	18.44	15.42	-3.02
Tamil	ta_lk	in-domain	64.30	69.06	4.76
Urdu	ur_pk	FLEURS	11.65	20.04	8.39
Urdu	ur_in	in-domain	20.17	17.35	-2.82
Urdu	ur_pk	in-domain	46.03	74.13	28.10

Table 5: CER of the best off-the-shelf baseline against the best LisTAya checkpoint on the same test set. Negative Δ favours the trained projector.

Language	Region	Matched	Mism.	Excess pp	Δ CER
English	Water	17.0	32.2	-15.2	7.39
Spanish (LatAm)	Water	2.1	1.2	0.9	0.07
French	Water	2.2	1.3	1.0	-0.43
Amharic	Earth	1.3	0.0	1.3	1.32
Swahili	Earth	1.3	0.0	1.3	-0.15
Hausa	Earth	1.3	0.0	1.3	-1.19
Indonesian	Water	2.0	0.1	1.9	-0.40
Urdu	Fire	3.4	1.3	2.2	1.44
Tamil	Fire	3.4	0.0	3.4	-14.70
Marathi	Fire	3.7	0.1	3.6	-0.44
Hindi	Fire	5.8	0.1	5.7	1.06

Table 6: Share of each decoder's post-training mix in the target language, from the Tiny Aya report. Excess is matched minus the mean of the two mismatched variants. English is the one language whose matched decoder saw *less* of it than the mismatched ones did.

Language	Test set	Whisper-M	Voxtral-M	Qwen2-A
Amharic	FLEURS	233.22	111.61	133.14
Amharic	in-domain	144.27	133.07	149.62
Seychellois Creole	in-domain	†	†	85.18
English	FLEURS	2.35	2.11	4.67
English	in-domain	10.39	9.95	16.80
Spanish (LatAm)	FLEURS	1.34	1.73	3.90
Spanish (LatAm)	in-domain	22.39	8.46	12.21
French	FLEURS	2.86	2.10	4.97
French	in-domain	10.93	12.06	15.19
Hausa	FLEURS	71.92	55.94	89.99
Hausa	in-domain	83.38	82.00	94.57
Hindi	FLEURS	15.64	5.74	93.96
Hindi	in-domain	43.22	24.50	101.29
Indonesian	FLEURS	3.26	4.08	31.30
Indonesian	in-domain	13.89	14.58	45.16
Marathi	FLEURS	60.25	19.45	97.39
Marathi	in-domain	38.12	18.62	96.27
Swahili	FLEURS	42.04	26.32	78.95
Swahili	in-domain	625.31	18.63	25.45
Tamil	FLEURS	16.22	48.02	109.62
Tamil	in-domain	18.44	30.27	105.29
Urdu	FLEURS	11.65	48.19	106.90
Urdu	in-domain	164.20	46.03	68.22

Table 7: CER of the off-the-shelf models on each language. Values above 100 indicate degenerate output rather than substitution errors; † marks a run that failed because the model does not accept the language. FLEURS has no Seychellois Creole, so that cell has no cross-domain row at all.

Language	**Variety**	**Status**	n	**CER**	Δ	**Ratio**
English	en_au	held out	3880	9.95	-0.93	0.91
English	en_jm	held out	117	7.97	-1.98	0.80
English	en_ke	held out	1822	25.02	9.54	1.96
English	en_nz	held out	3074	9.74	-1.48	0.85
English	en_pk	held out	237	87.70	77.31	7.89
English	en_sl	held out	1238	16.71	4.17	1.42
English	en_zm	held out	3527	17.77	5.88	1.59
Spanish (LatAm)	es_ar	held out	3032	20.22	10.14	1.83
Spanish (LatAm)	es_cl	held out	20912	23.35	11.14	2.30
Spanish (LatAm)	es_co	held out	1893	17.96	5.75	1.47
Spanish (LatAm)	es_es	held out	34140	7.96	-2.04	0.83
Spanish (LatAm)	es_pe	held out	4637	10.61	-1.91	0.81
Spanish (LatAm)	es_pr	held out	2742	13.92	1.08	1.13
Spanish (LatAm)	es_py	held out	1604	12.00	-0.21	0.98
Spanish (LatAm)	es_uy	held out	10730	21.41	11.73	1.96
French	fr_cd	held out	265	10.58	-3.75	0.70
French	fr_ci	held out	116	12.13	-2.62	0.83
Swahili	sw_tz	trained	10542	51.19	26.82	2.44
Tamil	ta_lk	held out	1224	88.76	45.86	2.93
Urdu	ur_in	trained	155	47.35	1.32	1.03

Table 8: Country varieties beyond each cell's training variety, median over the three off-the-shelf models, in CER. Δ and ratio are against the same model's in-domain point. Tanzanian Swahili and Indian Urdu were in their cells' training streams and are not held out.